

AUTONOMOUS RESCUE DRONE



Abstract

The Autonomous Search and Rescue Drone is a cutting-edge solution for emergency response. It uses RGB cameras to autonomously detect and locate survivors in real-time, navigating disaster areas with GPS-guided precision. The drone transmits live data to rescue teams, enabling rapid decisions and efficient operations in critical situations.



Motivation:

Natural disasters such as earthquakes, floods, and wildfires often create hazardous and inaccessible areas, making it difficult for rescue teams to locate survivors efficiently. In 2024, severe floods in Bangladesh affected over 5.8 million people and displaced more than 500,000. An Autonomous Search and Rescue Drone that can locate and identify survivors using real-time sensing and processing technologies and send Geotagged location data to rescue teams, which will reduce the time-consuming and human-risk-intensive nature of traditional SAR operations.



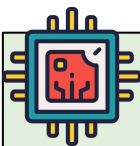
Objectives

- Rapid Response:** Reach disaster areas faster.
- Survivor Assistance:** Locate and deliver supplies to stranded individuals.
- Safety:** Minimize risks for human rescuers.



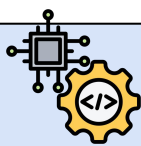
Key Features

- Survivor Detection:** RGB cameras locate survivors in various environments.
- Autonomous Navigation:** GPS-guided flight ensures precise search patterns.
- Real-Time Transmission:** Sends instant updates to rescue teams on the ground.
- Geo-Tagging:** Accurately maps survivor locations for rapid response.



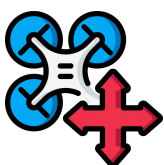
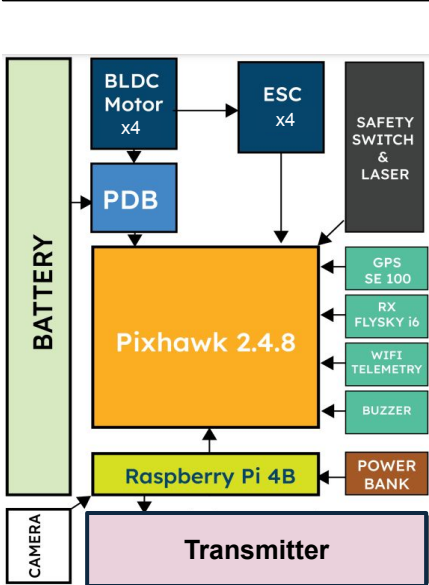
Hardware Integration

- Frame & Propulsion:** Quadcopter frame with brushless motors and ESCs.
- Flight Controller:** Integrates GPS and sensors for autonomous navigation.
- Sensors:** Camera for detection, GPS for tracking.
- Processing Unit:** Raspberry Pi 4B for real-time decision-making.
- Communication:** Telemetry module for real-time gps data transmission.

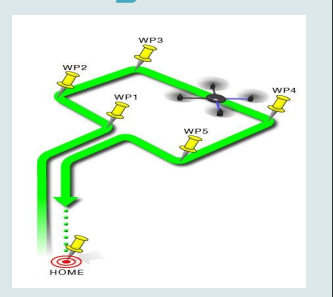


Firmware & Software

- Operating System:** Raspberry Pi OS for onboard system management and communication.
- Flight Software:** ArduPilot (Mission Planner) for autonomous GPS-based navigation.
- Survivor Detection:** OpenCV with YOLOv8n for visual data processing and survivor identification.



Autonomous Navigation

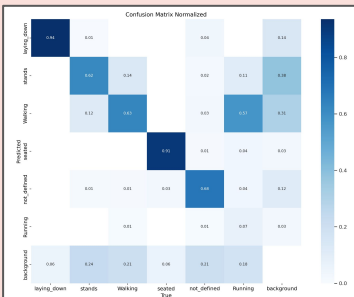


Human Detection

- Model Used:** Yolov8n ,Trained for 50 epochs
- Dataset:** SAR detection Dataset from Roboflow
- Train image:** 3000 ; **Test image:** 1000



- Matrices:**
- Precision :** 0.717
- Recall :** 0.6472
- mAP50 :** 0.68936
- mAP50-95 :** 0.40141



Impacts

- Scalable and Adaptable.
- Faster Response.
- Reduced Human Risk.
- Real-Time GPS - Data.
- Cost- Effective.
- Precision Navigation.
- Visual AI Detection.
- Enhanced Safety.



Future Works

- Power Efficiency.
- AI Model Enhancement.
- Real-Time Data Speed.
- Advanced Flight Stability.
- Obstacle Avoidance.
- Extended Testing.
- Cost Optimization.



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